

Chowdhury Abdulla Nasir

Tsukuba, Japan • abdullahnasirchowdhury1@gmail.com • [Portfolio](#) • [GitHub](#) • [LinkedIn](#)

Education

University of Tsukuba

Tsukuba, Japan

Graduate School of Systems and Information Engineering, Dept. of Computer Science
Master's Student (M1), Apr. 2026 – Present (*Research Student, Apr. 2025 – Mar. 2026*)
Advisor: Prof. Kazuhiro Fukui (福井和広), Computer Vision Laboratory (CVLab)

Leading University

Sylhet, Bangladesh

B.Sc. in Electrical and Electronic Engineering, 2019 – 2023
CGPA: 3.93 / 4.00 — 1st place, 26th batch (of 50 students)

Research & Work Experience

Computer Vision Laboratory (CVLab), University of Tsukuba

2025 – Present

Graduate Researcher, advised by Prof. Kazuhiro Fukui. Research on subspace representation methods for point cloud surface inspection under synthetic-to-real domain shift, and statistical robustness benchmarking of point cloud classification architectures.

Intex Research Lab

Sylhet, Bangladesh • 2024 – 2025

Research Assistant. Published a paper at a Scopus-indexed IEEE Xplore conference; annotated and labelled Bangla characters and words; benchmarked OCR engines for Bangla character recognition.

EARL Research Network

Sylhet, Bangladesh • 2021 – 2023

Research Assistant. Developed the lab website; published a paper at a Scopus-indexed IEEE Xplore conference; contributed to research projects and seminars.

Publications

- **Subspace Representation for Point Cloud Surface Inspection under Synthetic-to-Real Domain Shift.** ACCV 2026 (*under review*).
- **Do We Still Need Rotation-Invariant Architectures? A Statistical Robustness Benchmark for Point Cloud Classification.** WACV 2027, Datasets Track (*under review*).
- **Prototype and Simulation of a Real-Time GPS and PTS Architecture.** 2025 Int'l Conf. on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN), Rangpur, Bangladesh. IEEE Xplore. *Published*.
- **Performance Analysis of PyTesseract and EasyOCR for Bangla Optical Character Recognition on the Novel Bangla CrossHair Dataset.** 3rd Int'l Conf. on Intelligent Systems, Advanced Computing and Communication, 2025. IEEE Xplore (Scopus-indexed).
- **ALPR: ResNet50-powered Bangla License Plate Detection and OCR using Root Mean Square Prop Optimizer and Linear SVM Classifier.** IEEE 9th Int'l Conf. for Convergence in Technology (I2CT), 2024. IEEE Xplore (Scopus-indexed).

Selected Projects

Point Cloud Inspection API

2026

Unsupervised 3D defect detection: FPFH descriptors matched against a memory bank of defect-free parts via FAISS, served with FastAPI in Docker, benchmarked on MVTec 3D-AD. Requires no labelled defects.

Real-Time Detection & Tracking

2026

YOLO11 + ByteTrack pipeline for multi-object tracking, exported to PyTorch / ONNX / TensorRT with a measured CPU vs. GPU latency benchmark, plus a per-worker PPE (hardhat/vest) compliance demo. Deployed live on Google Cloud Run.

Bicol Cacao Field Tools

2026

Drip-irrigation pressure planner, P&ID water-system schematic, and Raspberry Pi controller design for smallholder cacao farms in Bicol, Philippines, built from field research with Team BEE, University of Tsukuba.

Skills

- **Languages:** Python, C/C++, MATLAB
- **ML / Computer Vision:** PyTorch, scikit-learn, Open3D, FAISS, OpenCV
- **GenAI / LLM:** LangChain, OpenAI & Hugging Face APIs
- **Systems & Tools:** FastAPI, Docker, Git, Linux, $\LaTeX$
- **Statistics:** Hypothesis testing, ANOVA, effect-size analysis, experimental design — applied in the WACV 2027 submission above

Certifications & Test Scores

- Machine Learning Specialization — Stanford University / DeepLearning.AI (Coursera)
- Data Scientist in Python Career Track — DataCamp
- IELTS: 8.0/9.0 (CEFR C1)
- Japanese: JLPT N4 (N3 exam scheduled Dec. 2026); conversational fluency

Positions of Responsibility

Website Maintainer, CVLab, University of Tsukuba

2025 – Present

References

Prof. Kazuhiro Fukui (福井和広) — Computer Vision Laboratory, Graduate School of Systems and Information Engineering, University of Tsukuba. *Contact available upon request.*